111

Pathophysiology and Classification of Lower Urinary Tract Dysfunction

Overview

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Keywords

bladder; pathophysiology; storage emptying; urethral sphincter

Questions

1. Which of the following best describes normal bladder behavior during the filling-storage phase of the micturition cycle?
    a. Low compliance due to elastic properties
    b. High compliance due to elastic properties
    c. Low compliance due to elastic and viscoelastic properties
    d. High compliance due to elastic and viscoelastic properties
    e. High compliance due to a low relaxation coefficient of the lamina propria
2. A patient who has significantly and urodynamically dangerous decreased compliance because of a replacement by collagen of other components of the stroma is generally best managed by:
    a. pharmacologic regimen.

b. hydraulic distention.
c. nerve section.
d. augmentation cystoplasty.
e. neuromodulation.

3. The “guarding reflex” refers to the:
   a. abrupt increase in striated sphincter activity seen with a cough during normal bladder filling/storage.
   b. spinal sympathetic inhibition of parasympathetic ganglion activity.
   c. gradual increase in striated sphincter activity seen during normal bladder filling/storage.
   d. gradual inhibition of the pontine-mesencephalic micturition center by the cerebral cortex during normal bladder filling/storage.
   e. gradual inhibition of the sacral spinal cord ventral nuclei by the pontine-mesencephalic brainstem during normal bladder filling/storage.
4. The primary effect of the spinal sympathetic reflexes that are evoked in animals during bladder filling and that facilitate bladder filling/storage is:
   a. neurally mediated stimulation of the α-adrenergic receptors in the area of the smooth sphincter.
   b. neurally mediated stimulation of the β-adrenergic receptors in the bladder body smooth musculature.
   c. direct inhibition of detrusor motor neurons in the sacral spinal cord.
   d. neurally mediated inhibition of cholinergic receptors in the area of the bladder body.
   e. neurally mediated sympathetic modulation of cholinergic ganglionic transmission.
5. The organizational center for the micturition reflex in an intact neural axis is the:
   a. pontine mesencephalic formation in the brainstem.
   b. frontal area of the cerebral cortex.
   c. parietal area of the cerebral cortex.
   d. cerebellum.

e. sacral spinal cord.

6. IVCs are most commonly seen in association with:
    a. sacral spinal cord neurologic disease or injury.
    b. infrasacral neurologic disease or injury.
    c. suprasacral neurologic disease or injury.
    d. peripheral nerve neurologic disease or injury.
    e. interstitial cystitis.
7. Using the functional classification system, the usual lower urinary tract dysfunction seen after a stroke would be categorized as:
    a. failure to store because of the bladder (overactivity).
    b. combined deficit (failure to store because of the bladder, failure to empty because of striated sphincter dyssynergy).
    c. combined deficit (failure to store because of the bladder, failure to empty because of a nonrelaxing outlet).
    d. failure to store because of the bladder (hypersensitivity).
    e. failure to store because of the outlet.
8. In the International Continence Society (ICS) classification system, the disorder described in question 7 would be characterized as:
    a. during storage, overactive neurogenic detrusor activity, increased sensation, low bladder capacity, and incompetent urethral closure mechanism; and during voiding, normal detrusor activity and abnormal urethral function (dysfunctional voiding).
    b. during storage, normal detrusor function, increased sensation, low bladder capacity, and normal urethral closure mechanism; and during voiding, normal detrusor activity and abnormal urethral function (dysfunctional voiding).
    c. during storage, overactive neurogenic detrusor activity, normal sensation, normal bladder capacity, and incompetent urethral closure mechanism; and during voiding, normal detrusor activity and normal urethral function.
    d. during storage, stable detrusor activity, reduced sensation, low bladder capacity, and normal urethral closure mechanism; and during emptying, normal detrusor activity and abnormal urethral function (dysfunctional voiding).

e. during storage, overactive neurogenic detrusor activity, normal sensation, low capacity, normal compliance, and normal urethral closure function; and during emptying, normal detrusor activity and normal urethral function.

9. In the current ICS terminology, “detrusor hyperreflexia” has been replaced by:
   a. detrusor instability.
   b. idiopathic detrusor overactivity.
   c. hyperactive bladder.
   d. neurogenic detrusor overactivity.
   e. neurogenic detrusor instability.
10. In the Krane-Siroky urodynamic classification system, a patient with post–cerebrovascular accident voiding dysfunction characterized by urgency, frequency, and urge incontinence would most commonly be characterized as having:
   a. detrusor areflexia, striated sphincter dyssynergia, and smooth sphincter dyssynergia.
   b. detrusor overactivty, striated sphincter synergia, and smooth sphincter dyssynergia.
   c. detrusor overactivty, striated sphincter dyssynergia, and smooth sphincter synergia.
   d. detrusor areflexia, striated sphincter synergia, and smooth sphincter dyssynergia.
   e. detrusor overactivty, striated sphincter synergia, and smooth sphincter dyssynergia.
11. In the Lapides classification system, a patient with post–cerebrovascular accident voiding dysfunction characterized by urgency, frequency, and urge incontinence would most commonly be characterized as having:
   a. sensory neurogenic bladder.
   b. motor paralytic bladder.
   c. uninhibited neurogenic bladder.
   d. reflex neurogenic bladder.
   e. autonomous neurogenic bladder.
12. A reflex neurogenic bladder, as described in the Lapides system classification, is characteristically seen in which of the following?

a. Traumatic spinal cord injury between the sacral spinal cord and the brainstem
b. Traumatic spinal cord injury between the sacral spinal cord and conus medullaris
c. Cerebrovascular accident and insulin-dependent diabetes mellitus
d. Non-insulin-dependent diabetes mellitus
e. Multiple sclerosis

13. In the Bors-Comarr system of classification, the term *unbalanced*, when applied to a patient with an upper motor neuron (UMN) lesion, implies:
    a. cerebellar lesion.
    b. involuntary bladder contractions during filling.
    c. areflexic bladder.
    d. decreased bladder compliance during filling.
    e. sphincter dyssynergia.

14 In the Bors-Comarr system, a patient with post–cerebrovascular accident voiding dysfunction characterized by urgency, frequency, and urge incontinence would most commonly be characterized as having:
    a. a UMN lesion, complete, and balanced.
    b. a UMN lesion, complete, and imbalanced.
    c. a lower motor neuron (LMN) lesion, complete, and imbalanced.
    d. an LMN lesion, incomplete, and balanced.
    e. a UMN lesion/LMN lesion, complete, and balanced.

15. Which of the following is an absolute requirement for a patient to be included in the symptom syndrome of overactive bladder?
    a. Nocturia
    b. Urinary frequency
    c. Urgency
    d. Urgency incontinence
    e. Detrusor overactivity

16. Which of the following pathophysiologic factors is shared by men and women with urinary incontinence (failure to store) due to outlet underactivity?

a. Bladder neck hypermobility
b. Intrinsic sphincter dysfunction
c. Proximal urethral hypermobility
d. Nonrelaxing striated sphincter
e. Bladder neck dysfunction

Answers

1. **d. High compliance due to elastic and viscoelastic properties.** The normal adult bladder response to filling at a physiologic rate is an almost imperceptible change in intravesical pressure. During at least the initial stages of bladder filling, after unfolding of the bladder wall from its collapsed state, this high compliance ($\Delta V/\Delta P$) of the bladder is due primarily to its elastic and viscoelastic properties. Elasticity allows the constituents of the bladder wall to stretch to a certain degree without any increase in tension. Viscoelasticity allows stretch to induce a rise in tension followed by a decay (stress relaxation) when the filling (stretch stimulus) slows or stops.
2. **d. Augmentation cystoplasty.** The viscoelastic properties of the stroma (bladder wall less smooth muscle and epithelium) and the urodynamically relaxed detrusor muscle account for the passive mechanical properties and normal bladder compliance seen during filling. The main components of stroma are collagen and elastin. When the collagen component increases, compliance decreases. This can occur with various types of injury, chronic inflammation, bladder outlet obstruction, and neurologic decentralization. Once decreased compliance occurs because of a replacement by collagen of other components of the stroma, it is generally unresponsive to pharmacologic manipulation, hydraulic distention, or nerve section. Most often, under those circumstances, augmentation cystoplasty is required to achieve satisfactory reservoir function.
3. **c. Gradual increase in striated sphincter activity seen during normal bladder filling/storage.** There is a gradual increase in urethral pressure during bladder filling, contributed to by at least

the striated sphincter element and perhaps by the smooth sphincteric element as well. The rise in urethral pressure seen during the filling/storage phase of micturition can be correlated with an increase in efferent pudendal nerve impulse frequency and in electromyographic activity of the periurethral striated musculature. This constitutes the efferent limb of a spinal somatic reflex, the so-called *guarding reflex*, which results in a gradual increase in striated sphincter activity during normal bladder filling and storage.

4. **e. Neurally mediated sympathetic modulation of cholinergic ganglionic transmission.** Does the nervous system affect the normal bladder response to filling? At a certain level of bladder filling, spinal sympathetic reflexes facilitatory to bladder filling/storage are clearly evoked in animals, a concept developed over the years by deGroat and colleagues, who have also cited indirect evidence to support such a role in humans. This inhibitory effect is thought to be mediated primarily by sympathetic modulation of cholinergic ganglionic transmission. Through this reflex mechanism, two other possibilities exist for promoting filling/storage. One is neurally mediated stimulation of the predominantly α-adrenergic receptors in the area of the smooth sphincter, the net result of which would be to cause an increase in resistance in that area. The second is neurally mediated stimulation of the predominantly β-adrenergic receptors (inhibitory) in the bladder body smooth musculature, which would cause a decrease in bladder wall tension. McGuire has also cited evidence for direct inhibition of detrusor motor neurons in the sacral spinal cord during bladder filling that is due to increased afferent pudendal nerve activity generated by receptors in the striated sphincter. Good evidence also seems to exist to support a tonic inhibitory effect of other neurotransmitters on the micturition reflex at various levels of the neural axis. Bladder filling and consequent wall distention may also release autocrine-like factors that influence contractility (e.g., nitric oxide, prostaglandins, peptides).
5. **a. Pontine mesencephalic formation in the brainstem.** Although the origin of the parasympathetic neural outflow to the bladder, the

pelvic nerve, is in the sacral spinal cord, the actual organizational center for the micturition reflex in an intact neural axis is in the brainstem, and the complete neural circuit for normal micturition includes the ascending and descending spinal cord pathways to and from this area and the facilitatory and inhibitory influences from other parts of the brain.

6. **c. Suprasacral neurologic disease or injury.** Involuntary contractions (IVCs) are most commonly seen associated with suprasacral neurologic disease or after suprasacral neurologic injury; however, they may also be associated with aging, inflammation or irritation of the bladder wall, bladder outlet obstruction, or stress urinary incontinence, or they may be idiopathic.
7. **a. Failure to store because of the bladder (overactivity).** The classic symptoms of post-stroke lower urinary tract dysfunction are urgency, frequency, and possible urgency incontinence. The urodynamic findings are generally detrusor overactivity (DO) during filling/storage with normal sensation and synergic sphincter activity during voluntary or involuntary emptying, unless the patient attempts to inhibit the DO with striated sphincter contraction. This translates simply in the functional system to a failure to store because of the bladder.
8. **e. During storage, overactive neurogenic detrusor activity, normal sensation, low capacity, normal compliance, and normal urethral closure function; and during emptying, normal detrusor activity and normal urethral function.** The micturition dysfunction of a stroke patient with urgency incontinence would most likely be classified during storage as overactive neurogenic detrusor function, normal sensation, low capacity, normal compliance, and normal urethral closure function. During voiding, the dysfunction would be classified as normal detrusor activity and normal urethral function, assuming that no anatomic obstruction existed.
9. **d. Neurogenic detrusor overactivity.** The Standardization Subcommittee of the ICS made some changes in definitions of terms (published as a committee report in 2002). One change was

to eliminate the terms *detrusor hyperreflexia* and *instability* and replace them with the terms *neurogenic detrusor overactivity* and *idiopathic detrusor overactivity* (Abrams P et al., 2003). [a]

10. **b. Detrusor overactivty striated sphincter synergia, and smooth sphincter dyssynergia.** When exact urodynamic classification is possible, this system provides a truly precise description of the voiding dysfunction that occurs. If a normal or hyperreflexic detrusor exists with coordinated smooth and striated sphincter function and without anatomic obstruction, normal bladder emptying should occur. *Detrusor overactivty* is most commonly associated with neurologic lesions above the sacral spinal cord. Striated sphincter dyssynergia is most common after complete suprasacral spinal cord injury, following the period of spinal shock. Smooth sphincter dyssynergia is seen most classically in autonomic overactivty, when it is characteristically associated with detrusor overactivty and striated sphincter dyssynergia. *Detrusor areflexia* may be secondary to bladder muscle decompensation or to various other conditions that produce inhibition at the level of the brainstem micturition center, sacral spinal cord, bladder ganglia, or bladder smooth muscle. Patients with a voiding dysfunction secondary to detrusor areflexia generally attempt bladder emptying by abdominal straining, and their continence status and the efficiency of their emptying efforts are determined by the status of their smooth and striated sphincter mechanisms.
11. **c. Uninhibited neurogenic bladder.** Lapides contributed significantly to the classification and care of the patient with neuropathic voiding dysfunction by slightly modifying and popularizing a system originally proposed by McLellan in 1939. Lapides' classification differs from that of McLellan in only one respect, and that is the division of the group "atonic neurogenic bladder" into sensory neurogenic bladder and motor neurogenic bladder. This remains one of the most familiar systems to urologists and nonurologists because it describes in recognizable shorthand

the clinical and cystometric conditions of many types of neurogenic voiding dysfunction. An uninhibited neurogenic bladder was described originally as resulting from injury or disease to the "corticoregulatory tract." The sacral spinal cord was presumed to be the micturition reflex center, and this corticoregulatory tract was believed to normally exert an inhibitory influence on the sacral micturition reflex center. A destructive lesion in this tract would then result in overfacilitation of the micturition reflex. Cerebrovascular accident, brain or spinal cord tumor, Parkinson disease, and demyelinating disease were listed as the most common causes in this category. The voiding dysfunction is most often characterized symptomatically by frequency, urgency, and urge incontinence, as well as urodynamically by normal sensation with DO at low filling volumes. Residual urine is characteristically low unless anatomic outlet obstruction or true smooth or striated sphincter dyssynergia occurs. The patient can generally initiate a bladder contraction voluntarily but is often unable to do so during cystometry because sufficient urine storage cannot occur before the DO is stimulated.

12. **a. Traumatic spinal cord injury between the sacral spinal cord and the brainstem.** Reflex neurogenic bladder describes the post-spinal shock condition that exists after complete interruption of the sensory and motor pathways between the sacral spinal cord and the brainstem. Most commonly, this occurs in traumatic spinal cord injury and transverse myelitis, but it may occur with extensive demyelinating disease or any process that produces significant spinal cord destruction as well. Typically, there is no bladder sensation and there is inability to initiate voluntary micturition. Incontinence without sensation generally results because of low-volume DO. Striated sphincter dyssynergia is the rule. This type of lesion is essentially equivalent to a complete UMN lesion in the Bors-Comarr system.
13. **e. Sphincter dyssynergia.** This system applies only to patients with neurologic dysfunction and considers three factors: (1) the anatomic localization of the lesion; (2) the neurologic completeness or incompleteness of the lesion; and (3) a designation as to whether

lower urinary tract function is balanced or unbalanced. The latter terms are based solely on the percentage of residual urine relative to bladder capacity. Unbalanced signifies the presence of greater than 20% residual urine in a patient with a UMN lesion or 10% in a patient with an LMN lesion. This relative residual urine volume was ideally meant to imply coordination (synergy) or dyssynergia between the smooth and striated sphincters of the outlet and the bladder, during bladder contraction or during attempted micturition by abdominal straining or the Credé method.

14. **a. A UMN lesion, complete, and balanced.** In this system, UMN bladder refers to the pattern of micturition that results from an injury to the suprasacral spinal cord after the period of spinal shock has passed, assuming that the sacral spinal cord and the sacral nerve roots are intact and that the pelvic and pudendal nerve reflexes are intact. LMN bladder refers to the pattern resulting if the sacral spinal cord or sacral roots are damaged and the reflex pattern through the autonomic and somatic nerves that emanate from these segments is absent. This system implies that if skeletal muscle spasticity exists be low the level of the lesion, the lesion is above the sacral spinal cord and is by definition a UMN lesion. This type of lesion is characterized by DO during filling. If flaccidity of the skeletal musculature below the level of a lesion exists, an LMN lesion is assumed to exist, implying detrusor areflexia. Exceptions occur and are classified in a mixed lesion group characterized either by DO with a flaccid paralysis below the level of the lesion or by detrusor areflexia with spasticity or normal skeletal muscle tone neurologically below the lesion level. UMN lesion, complete, and imbalanced implies a neurologically complete lesion above the level of the sacral spinal cord that results in skeletal muscle spasticity below the level of the injury. DO occurs during filling, but a residual urine volume of greater than 20% of the bladder capacity is left after bladder contraction, implying obstruction in the area of the bladder outlet during the involuntary detrusor contraction. This obstruction is generally due to striated sphincter dyssynergia, typically occurring in patients who are paraplegic and quadriplegic with lesions between the cervical and the sacral spinal

cord. Smooth sphincter dyssynergia may be seen as well in patients with lesions above the level of T6, usually in association with autonomic DO. LMN lesion, complete, and imbalanced implies a neurologically complete lesion at the level of the sacral spinal cord or of the sacral roots, resulting in skeletal muscle flaccidity below that level. Detrusor areflexia results, and whatever measures the patient may use to increase intravesical pressure during attempted voiding are not sufficient to decrease residual urine to less than 10% of bladder capacity.

15. **c. Urgency.** Overactive bladder is defined (ICS) as urgency, with or without urinary urgency incontinence, usually with frequency and nocturia. One-third of the patients have incontinence, but two-thirds do not. Frequency and nocturia are usually but not always present. Detrusor overactivity is a urodynamic term indicating an involuntary bladder contraction. Urgency may or may not be associated with detrusor overactivity on a urodynamic study.
16. **b. Intrinsic sphincter dysfunction.** Failure to store because of outlet underactivity in the female is due to a combination of a failure of support, generally accompanied by hypermobility of the bladder outlet and intrinsic sphincter dysfunction (ISD). **It is impossible to have effort-related incontinence in the woman without some element of ISD.** Outlet-related incontinence in the male is most commonly seen after prostatectomy, and there is no pathophysiologic factor of hypermobility involved. The condition is essentially ISD. A nonrelaxing striated sphincter would not produce urinary incontinence, nor would bladder neck dysfunction.

Chapter Review

1. The spinal sympathetic reflexes are facilitatory to bladder filling/storage.
2. The micturition cycle is divided into two phases: (1) bladder filling/urine storage and (2) bladder empting/voiding.
3. There are two urethral sphincters: (1) the smooth urethral sphincter is the smooth musculature of the bladder neck and proximal urethra;

the smooth sphincter is not under voluntary control; and (2) the striated sphincter, which has two parts—the striated intramural sphincter, called the rhabdosphincter, and the external striated sphincter, which is part of the levator musculature. This sphincter is under voluntary control.

4. While collagen content of the bladder wall increases, compliance decreases.
5. There is increased afferent input when inflammation or irritation occurs, causing hypersensitivity to pain.
6. The hammock hypothesis of continence proposes that there is a fixed dorsal portion of the urethra due to fascial attachments against which the ventral aspect of the urethra is compressed.
7. Increases in intra-abdominal pressure are transmitted to the proximal urethra (as well as the mid-urethra in females).
8. The bladder response to filling at a physiologic rate is an almost imperceptible change in intravesical pressure.
9. Elasticity allows the constituents of the bladder wall to stretch to a certain degree without any increase in tension. Viscoelasticity allows stretch to induce a rise in tension followed by a decay (stress relaxation) when the filling (stretch stimulus) slows or stops.
10. IVCs are most commonly seen associated with suprasacral neurologic disease or after suprasacral neurologic injury; however, they may also be associated with aging, inflammation, or irritation of the bladder wall, bladder outlet obstruction, or stress urinary incontinence, or they may be idiopathic.
11. Overactive bladder is defined (ICS) as urgency, with or without urinary urgency incontinence, usually with frequency and nocturia.
12. It is impossible to have effort-related incontinence in the woman without some element of ISD.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition*, on the Expert Consult website.

112

Evaluation and Management of Women With Urinary Incontinence and Pelvic Prolapse

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Keywords

cystoscopy; pelvic floors disorders; pelvic organ prolapse; POP-Q; urinary incontinence; urodynamics

Questions

1. Which of the following is *not* a risk factor for urinary incontinence?
    a. Increased BMI
    b. Male gender
    c. Fecal incontinence
    d. Smoking
    e. Increased age
2. A 57-year-old woman is advised to start on estrogen replacement therapy. She should be told that:
    a. past history of birth control pill or systemic hormone use can increase the risk of stress incontinence.
    b. active systemic exogenous estrogen can decrease stress incontinence.
    c. low endogenous estrogen level is a risk for the development of stress incontinence.

d. hormone replacement has no established effect on continence.
e. systemic estrogen therapy is preferential to vaginal therapy for incontinence.

3. Patients with vaginal vault prolapse and no symptoms of stress urinary incontinence should undergo a sacrocolpopexy only (i.e., no sling):
   a. regardless of preoperative testing.
   b. if stress urinary incontinence was only demonstrated on physical examination.
   c. if no stress urinary incontinence was demonstrated on either physical examination or urodynamic testing.
   d. if stress urinary incontinence was demonstrated only during urodynamic evaluation.
   e. if they had a previous history of a bulking agent.
4. A 49-year-old woman has prolapse that is recorded as Aa-3, Ba-3, Ap-3, Bp-3. She has:
   a. no prolapse.
   b. stage 1 prolapse.
   c. stage 2 prolapse.
   d. stage 3 prolapse.
   e. stage 4 prolapse.
5. A patient is suspected of having a urinary tract–vaginal fistula. Which is the best dye test method to facilitate diagnosis of an isolated ureterovaginal fistula?
   a. Oral pyridium.
   b. Intravesical indigo carmine.
   c. Intravenous indigo carmine.
   d. Simultaneous oral pyridium and intravenous indigo carmine.
   e. Simultaneous oral pyridium and intravesical indigo carmine.
6. A healthy 65-year-old woman has bothersome prolapse and stress urinary incontinence (SUI). Exam shows stage 3 anterior prolapse and no apical or posterior prolapse or stress incontinence. On urodynamics she has a stable bladder and no SUI. The next step is repeat stress testing with:
   a. the urethral catheter removed.
   b. the prolapse reduced.

c. the prolapse reduced and the urethral catheter removed.
d. both the rectal and urethral catheters removed.
e. a smaller-caliber urethral catheter.

7. The Pelvic Organ Prolapse Quantification (POP-Q) system:
   a. is a simple six-point quantification system for pelvic prolapse.
   b. was created in an effort to quantify pelvic organ prolapse and urinary incontinence.
   c. includes six specific points of position measurement in relation to the introitus.
   d. includes a simplified five-level staging system that does not require listing each of the points specifically.
   e. includes measurement of the total vaginal length performed *without* reduction of the prolapse.
8. A 34-year-old female is undergoing urodynamics for symptoms of urinary hesitancy. She is noted to have an absence of electromyography (EMG) recruitment with a squeezing of the clitoris. This represents a:
   a. disruption at the level of sacral nerve roots 2 to 4.
   b. dysfunction in the cauda equina.
   c. positive bulbocavernosus reflex (BCR).
   d. normal finding in 30% of normal females.
   e. problem with technique in assessing for the BCR.
9. The Food and Drug Administration (FDA) issued a mandate ordering the manufacturers to stop the sale and distribution of all remaining surgical mesh indicated for:
   a. slings.
   b. transvaginal prolapse repairs.
   c. transabdominal prolapse repairs.
   d. all prolapse repairs.
   e. all slings and pelvic prolapse repairs.
10. A 62-year-old man has bothersome urinary frequency and urgency associated with rare leakage. He has no obstructive symptoms, and postvoid residual (PVR) is 20 mL. Urine analysis (UA) is negative. The only therapy he has tried is a 2-week course of oxybutynin, but he is currently on no medications. The next step is:

a. behavioral therapy, including dietary modification, fluid management, and bladder training.
b. alpha-blocker therapy.
c. repeat a trial of antimuscarinic therapy.
d. beta-3 agonist therapy.
e. combination antimuscarinic and beta-3 agonist therapy.

11. “Eyeball urodynamics” can provide information regarding:
    a. detrusor compliance.
    b. bladder outlet obstruction.
    c. abdominal leak point pressure.
    d. detrusor leak point pressure.
    e. detrusor-sphincter coordination.
12. Multichannel urodynamics:
    a. is the most accurate diagnostic tool available for the evaluation of incontinence.
    b. should be used in all patients with incontinence.
    c. includes three directly measured values: detrusor pressure (Pdet), vesical pressure (Pves), and abdominal pressure (Pabd).
    d. is not helpful in determining if a patient is at risk of developing upper tract deterioration.
    e. all of the above.
13. Urodynamics should be considered:
    a. when conservative measures have failed.
    b. when the clinical picture is unclear.
    c. when the symptoms cannot be confirmed by the clinician.
    d. in patients who have undergone previous pelvic floor reconstruction.
    e. all of the above.
14. Electromyelogram should:
    a. be performed on all patients undergoing urodynamics.
    b. demonstrate recruitment during the filling phase.
    c. be active during coughing.
    d. be silent with bulbocavernosus reflex (BCR).
    e. be active during the voiding phase.

15. A 50-year-old lady presents with pure stress urinary incontinence. She has no other lower urinary tract symptoms (LUTS) and no previous pelvic surgeries. According to the American Urological Association/Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction (AUA/SUFU) SUI guideline, the initial evaluation of this patient should include a thorough history with assessment of symptoms and the following:
    a. Physical and pelvic exam with demonstration of SUI, urinalysis, postvoid residual, cystoscopy.
    b. Physical and pelvic exam with demonstration of SUI, urinalysis, postvoid residual, cystoscopy, urodynamics.
    c. Physical and pelvic exam with demonstration of SUI, urinalysis, cystoscopy, urodynamics.
    d. Physical and pelvic exam with demonstration of SUI, urinalysis, postvoid residual.
    e. Physical and pelvic exam with demonstration of SUI, urinalysis, postvoid residual, urodynamics.

Answers

1. **b. Male gender.** Obesity, advanced age, female gender, smoking, and associated pelvic floor disorders are risk factors for the development of urinary incontinence.
2. **a. Past history of birth control pill or systemic hormone use can increase the risk of stress incontinence.** Evidence shows that the use of exogenous hormones, currently or in the past, can increase the risk of development or exacerbation of existing stress incontinence such that local hormone replacement therapy is no longer considered part of the routine treatment for stress incontinence. However, local hormone replacement therapy can be helpful in the treatment of urgency incontinence. Vaginal hormone replacement is preferred to systemic therapy for treatment of lower urinary tract symptoms. Low endogenous hormone levels do not increase the risk of stress incontinence.

3. c. **If no stress urinary incontinence was demonstrated on either physical examination or urodynamic testing.** According to the AUA guidelines on surgical treatment of female stress urinary incontinence, a physician should demonstrate objective evidence of stress urinary incontinence prior to considering surgical management. Stress urinary incontinence could be demonstrated on physical exam or urodynamic testing. Based on the CARE and OPUS trials, patients with objective evidence of stress urinary incontinence should be counseled to consider concomitant surgical management of stress urinary incontinence. Patients with previous history of anti-incontinence procedure and objective evidence of stress urinary incontinence should still be counseled on surgical management of incontinence. If there is no objective evidence of stress urinary incontinence, one may consider not proceeding with surgical management of incontinence at time of prolapse repair.
4. **a. No prolapse.** The Pelvic Organ Prolapse-Quantification ("POP-Q") system is a 9-measure system that was created in an effort to provide objectivity to POP quantification. Six vaginal points labeled Aa, Ba, C, D, Ap, and Bp are measured in relation to the hymenal ring during a Valsalva maneuver. Aa and Ba represent points along the anterior vaginal wall, while Ap and Bp represent the posterior wall. Point C is the most distal point of the vaginal cuff or cervix, and D is the distance to the posterior fornix and is measured only if the cervix is still present. Points above the hymen are considered negative and points below the hymen are positive with a maximal established range of −3 to +3. The remaining three points are the genital hiatus (gh), which represents the size of the vaginal opening; the perineal body (pb), which represents the distance between the vagina and the anus; and the total vaginal length (tvl), which is measured by reducing the prolapse and measuring the depth of the vagina. A staging system based upon the above measurements ranges from stage 0 (no prolapse), in which all A and B points are −3, to stage 4, in which the leading edge is an absolute number that is no less than tvl −<2 cm (and is preceded by a "+" since it is below the hymen).

5. **a. Oral pyridium.** Oral pyridium will be excreted in the urine, so it will discolor a vaginal tampon whether the fistula is vesicovaginal or ureterovaginal or both. Intravenous indigo carmine would also demonstrate tampon staining in a similar pattern, but this is not practical as it involves the need for intravenous access. Presumably, staining higher on the tampon would indicate a ureterovaginal fistula, while lower staining would suggest a vesicovaginal fistula, but it can be difficult to differentiate with certainty as the dye can diffuse and 12% of patients with a vesicovaginal fistula have a concomitant ureteral fistula. Direct placement of dye (i.e., indigo carmine or methylene blue) into the bladder would only discolor the tampon if there is involvement of the bladder, but an isolated ureterovaginal fistula would be missed with this method. The best dye administration to evaluate for an isolated ureterovaginal fistula would be simultaneous intravesical blue dye and oral pyridium that would reveal orange staining only.
6. **b. The prolapse reduced.** When a patient who has subjective SUI and high-grade prolapse does not elicit SUI on urodynamics (UDS) studies, the AUA/SUFU urodynamics guidelines recommend reduction of the prolapse and repeat stress testing. In patients who have SUI with no prolapse, the next step is to repeat stress testing with the urethral catheter removed. In this situation, one would first repeat stress testing with the prolapse adequately reduced followed by removal of the urethral catheter. While downsizing the catheter or removing all catheters might result in demonstration of the leakage, this is not necessary or recommended by guidelines.
7. **d. Includes a simplified five-level staging system that does not require listing each of the points specifically.** POP-Q is a nine-point system that was created to objectify the assessment of pelvic organ prolapse. It measures six specific points in the vagina in relation to the hymen. The remaining three points include the total vaginal length, measured with the vagina completely reduced, the perineal body, and the genital hiatus. It has been simplified into to a five-stage system that does not require specific listing of each of the nine points.

8. **d. Normal finding in 30% of normal females.** BCR represents S2-4 and is present in all normal males and 70% of normal females. It is elicited by squeezing the glans penis or clitoris. During urodynamics, a positive is represented by increased EMG activity.
9. **b. Transvaginal prolapse repairs.** In 2019, the FDA released a mandate ordering the manufacturers to halt the sale and distribution of mesh placed transvaginally specifically for the repair of pelvic prolapse (https://www.fda.gov/news-events/press-announcements/fda-takes-action-protect-womens-health-orders-manufacturers-surgical-mesh-intended-transvaginal). Although the mandate excluded slings and transabdominally placed mesh for prolapse repair, unfortunately, subsequent media communication regarding mesh litigation created patient confusion and concern about the use of mesh in the pelvic floor in general. This prompted a response from the Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) (https://sufuorg.com/about/news/message-about-fda-statement-on-mus.aspx).
10. **a. Behavioral therapy, including dietary modification, fluid management, and bladder training.** Technically, this patient has not yet tried the first line of therapy according to the OAB guidelines. The best response is behavioral therapy that entails bladder exercises to train the bladder to overcome the sense of urgency when it occurs, pelvic floor exercises, fluid management, and avoidance of bladder irritants. Responses c, d, and e are all reasonable options to try in conjunction with behavioral therapy. Alpha-blockade would not be helpful in this situation in which the patient has no obstructive symptoms.
11. **a. Detrusor compliance.** "Eyeball UDS" is a simple alternative to full multichannel UDS that can provide ample information in selected patients. The study can determine bladder sensation, compliance, stability, and capacity as well as outlet competence and PVR. A 60-mL catheter tip syringe with the barrel removed is placed into the end of the catheter through which the bladder is filled by gravity. The height of the meniscus above the bladder represents the intravesical pressure. During the filling phase, a rise

and fall in the meniscus may represent bladder overactivity, whereas a consistent gradual rise suggests compromised detrusor compliance. Because of the absence of the abdominal pressure (Pabd) channel, no information about the contribution of abdominal pressure is gleaned, and it is therefore not possible to definitively establish the presence or absence of significant outlet obstruction. Similarly, there is no information about urethral function or EMG afforded by this method, thereby making c, d, and e incorrect responses.

12. **a. Is the most accurate diagnostic tool available for the evaluation of incontinence.** Multichannel urodynamics is currently the most accurate diagnostic tool available for the evaluation of urinary incontinence. Whether it is necessary in the assessment of all patients with urinary incontinence remains controversial. Findings on urodynamics, which include direct measurements of vesical and abdominal pressures (Pves and Pabd, respectively) and a calculated measure of detrusor pressure (Pdet), can provide helpful information including findings such as elevated Pdet that may suggest a patient is at increased risk of developing upper deterioration.
13. **e. All of the above.** While a clinician may reasonably forego performing UDS on the index patient with SUI and no urinary urgency, the guidelines state that UDS may be considered in patients who are considering undergoing invasive, potentially morbid, or irreversible treatments. This is left to the clinician, and the philosophy is that if the study may answer an unanswered question or somehow change the course of care, it may be considered. However, any patient with a picture complicated by issues such as previous pelvic or anti-incontinence surgery, radiation therapy, neurologic disease, or difficult or unclear diagnosis should be considered for urodynamics.
14. **c. Be active during coughing.** EMG should be used in selected patients. Recruitment should occur with BCR in all men and 70% of women and in the face of increased intra-abdominal pressure, such as with coughing. It should be silent during voiding to allow passage of urine without outlet resistance.

15. **d. Physical and pelvic exam with demonstration of SUI, urinalysis, postvoid residual.** According to AUA/SUFU guidelines, the initial evaluation of patients with SUI desiring to undergo surgical intervention should include the following: history with assessment of symptoms, physical examination (including pelvic exam), objective demonstration of SUI, assessment of postvoid residual, and urinalysis. Additional evaluations such as cystoscopy and urodynamics should be limited to patients who cannot be undiagnosed based on initial evaluation or who demonstrate SUI, neurogenic bladder, abnormal urinalysis, urgency-predominant mixed urinary incontinence, elevated PVR, high-grade prolapse, and evidence of significant voiding dysfunction.

Chapter Review

1. Pelvic floor disorders (PFDs) are a prevalent worldwide health concern.
2. Careful history and physical with objective demonstration of incontinence are integral in the proper evaluation of patients with PFDs.
3. Supplementary evaluation including urinalysis, postvoid residual, urodynamics, endoscopy, and radiologic imaging can be helpful in facilitating a complete and accurate diagnosis.
4. A variety of surgical and nonsurgical treatment options are available for treatment of urinary incontinence and pelvic organ prolapse.
5. The severity of PFD symptoms should be taken into account when considering treatment.
6. A thorough history is essential in the diagnostic evaluation of patients with PFDs.
7. Queries specific to the character, severity, duration, and quantity of incontinence and other symptoms related to pelvic floor function should be performed.
8. Attention should be paid to the impact of PFD symptoms on quality of life (QoL).

9. When appropriate, the clinician should present questions specific to the female with potential pelvic organ prolapse and the male with potential prostate issues.
10. Queries regarding past medical and surgical history, obstetric and gynecologic history, radiation therapy, trauma, and medications may provide important information.
11. A properly performed physical examination is imperative in the evaluation of patients with PFDs.
12. The Centers for Medicare/Medicaid require specific elements of both male and female genitourinary examination in order to meet coding guidelines (see Table 112.4).
13. Assessment of pelvic organ prolapse (POP) should ideally be performed in both the supine and standing positions.
14. Several classification and quantification systems are available for assessment of POP, the most widely used of which (Baden-Walker and POP-Q) are illustrated in Figs. 112.1 and 112.2.
15. A neurological examination and rectal examination should be performed in appropriate patients in order to obtain complete clinical information important in the assessment of lower urinary tract and pelvic floor function.
16. It has been demonstrated that patient subjective history alone often does not reflect an accurate or complete picture of their symptomatology complex.
17. Several instruments designed to facilitate symptom quantification have been developed and include tools such as voiding diaries, symptom and QoL questionnaires, and pad tests.
18. Voiding diaries can be both diagnostic and therapeutic, as they can provide patients with insights into behaviors that may be contributing to their voiding symptoms.
19. Pad tests may be helpful, particularly in the academic setting, to quantify incontinence symptoms. Loss of up to 8 g of urine in 24 hours may be considered normal, although the International Consultation on Incontinence (ICI) considers loss of greater than 1.3 g to be a positive 24-hour test.
20. Validated questionnaires are available to assess symptoms and quality of life in patients with PFDs.

21. Urodynamics (UDS) is the most accurate tool available for the assessment of LUT function and provides information regarding urine storage and emptying as they are affected by the bladder and the bladder outlet.
22. Although the use of routine UDS for straightforward incontinence is a topic of discussion, UDS should be strongly considered prior to intervention in patients who have a complex clinical picture due to failed previous treatment or surgery, mixed incontinence, obstructive symptoms, significantly elevated PVR, neurologic disease, or other medical conditions that may contribute to the LUT function, such as diabetes mellitus, pelvic prolapse, or history of radiation therapy.
23. "Eyeball UDS" may provide an approximate picture of the bladder capacity, sensation, stability, compliance, and outlet resistance when formal UDS is not available.

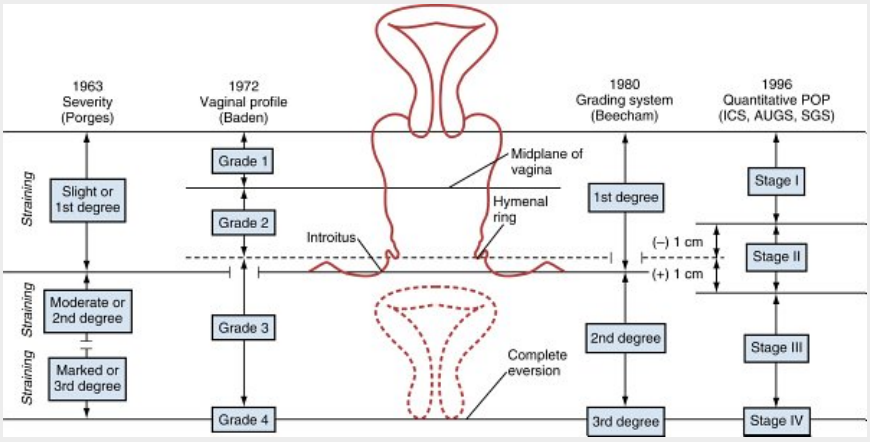
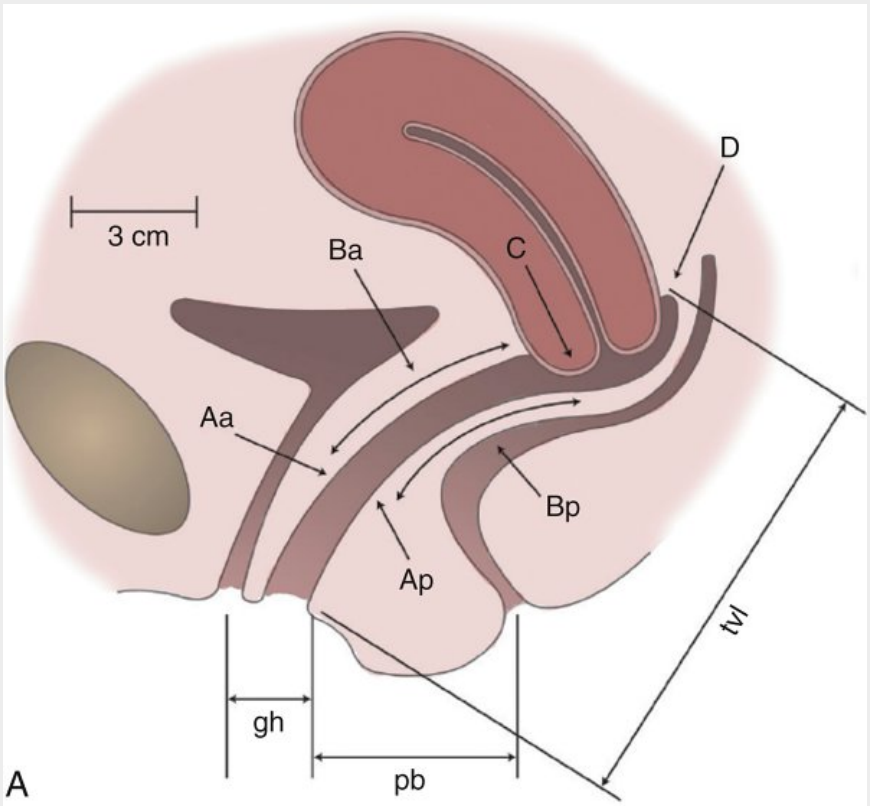


FIG. 112.1 Visual comparison of systems used to quantify pelvic organ prolapse (POP). *AUGS*, American Urogynecologic Society; *ICS*, International Continence Society; SGS, Society of Gynecologic Surgeons.
From Theofrastous JP, Swift SE: The clinical evaluation of pelvic floor dysfunction. *Obstet Gynecol Clin North Am* 25:783–804, 1998.



A

Point	Description	Range of values
Aa	Anterior vaginal wall 3 cm proximal to the hymen	–3 cm to +3 cm
Ba	Most distal position of remaining upper anterior vaginal wall	–3 cm to +tvl
C	Most distal edge of cervix or vaginal cuff scar	–
D	Posterior fornix (N/A if posthysterectomy)	–
Ap	Posterior vaginal wall 3 cm proximal to the hymen	–3 cm to +3 cm
Bp	Most distal position of remaining upper posterior vaginal wall	–3 cm to +tvl
gh (genital hiatus)	Measured from middle of external urethral meatus to posterior midline hymen	–
pb (perineal body)	Measured from posterior margin of gh to middle of anal opening	–
tvl (total vaginal length)	Depth of vagina when point D or C is reduced to normal position	–

B

FIG. 112.2 (A) Landmarks for the POP-Q system. (B) POP-Q points of reference. A, From Bump RC, Mattiasson A, Bo K, et al: The standardization of terminology of female pelvic organ prolapse and pelvic floor dysfunction. *Am J Obstet Gynecol* 175:10–17, 1996.

24. Multichannel UDS offers extensive evaluation of LUT function. It involves direct measurement of bladder and intra-abdominal pressure (Pves and Pabd, respectively) and a calculated assessment of detrusor pressure that is independent of abdominal pressure (Pdet). Pves-Pabd=Pdet.
25. Fluoroscopic imaging provides useful adjunctive information, such as the position and status of the bladder base and bladder neck, the presence of vesicoureteral reflux, and direct visualization of urinary leakage under real-time circumstances. Video imaging should be considered when the diagnosis cannot be made with certainty without simultaneous understanding of the anatomy in conjunction with the functional findings.

113

Evaluation and Management of Men With Urinary Incontinence

Riyad Tasher Al Mousa, and Hashim Hashim

Keywords

antimuscarinics; artificial urinary sphincter; botox; male slings; sacral neuromodulation; stress urinary incontinence; urinary incontinence; urgency urinary incontinence

Questions

1. Which of the following is the definition of stress urinary incontinence (UI)?
    a. Leakage of urine with urgency
    b. Leakage of urine with an increase in intra-abdominal pressure
    c. Leakage of urine while asleep
    d. Leakage of urine with urgency and effort
    e. Leakage of urine without being aware of it
2. Which of the following symptoms is not considered a bladder storage symptom?
    a. Frequency
    b. Urgency
    c. Hesitancy
    d. Nocturia
    e. UI
3. Which of the following is not a potential risk factor for UI in men?

a. Age
b. Race
c. Urological infection
d. Cognitive impairment
e. Prostatectomy

4. Initial assessment of men with UI includes all of the following except:
    a. flow test.
    b. invasive urodynamics.
    c. frequency/volume chart.
    d. urinalysis.
    e. quality-of-life questionnaire.
5. Which of the following is not a treatment for urgency urinary incontinence (UUI)?
    a. Bladder training
    b. Pelvic floor muscle training
    c. Antimuscarinics
    d. Duloxetine
    e. Botox
6. Which of the following is not a treatment for stress urinary incontinence (SUI) in men?
    a. Pelvic floor muscle training
    b. Penile clamp
    c. Antimuscarinics
    d. Male sling
    e. Artificial urinary sphincter
7. Which of the following is not measured by the International Prostate Symptom Score (AUA-SI)?
    a. Urgency
    b. Frequency
    c. Nocturia
    d. Incontinence
    e. Straining

Answers

1. **b. Leakage of urine with an increase in intra-abdominal presssure.** Leakage of urine with urgency is known as UUI. Leakage of urine while asleep is nocturnal enuresis. Leakage of urine on urgency and effort is mixed UI, and leakage of urine without being aware of it is insensible UI.
2. **c. Hesitancy.** Frequency, urgency, nocturia, and urinary incontinence are all bladder storage symptoms, while hesitancy is a voiding symptom.
3. **b. Race.** Potential risk factors for UI in men include increased age, urological infection, neurological and cognitive impairment, prostatectomy, and drugs. Ethnicity and race were not shown to have any statistical significance as a potential risk factor for UI in men.
4. **b. Invasive urodynamics.** Men with UI should be assessed with noninvasive baseline investigations including a flow test, frequency/volume chart or bladder diary, urinalysis to exclude infection and blood in the urine, and a quality-of-life questionnaire to assess the impact of the incontinence on quality of life. Invasive urodynamics, including filling cystometry and pressure flow studies, is reserved for use following failure of conservative and medical therapies and when it will alter the management of the patient.
5. **d. Duloxetine** Bladder training, pelvic floor muscle training, and antimuscarinics are first-line treatments of patients with UUI. If these fail, then patients can be treated with cystoscopic intradetrusor botulinum toxin-A injections. Duloxetine is a serotonin, norepinephrine reuptake inhibitor that has been licensed for the treatment of SUI in women but not for the treatment of UUI.
6. **c. Antimuscarinics.** Antimuscarinics are licensed for the treatment of overactive bladder syndrome and not SUI.
7. **d. Incontinence.** The IPSS or AUA-SI assesses three storage symptoms (urgency, frequency, nocturia), three voiding symptoms (intermittency, weak stream, straining), and one post-micturition symptom (incomplete emptying). It does not assess for UI (see Fig. 113.1AB).

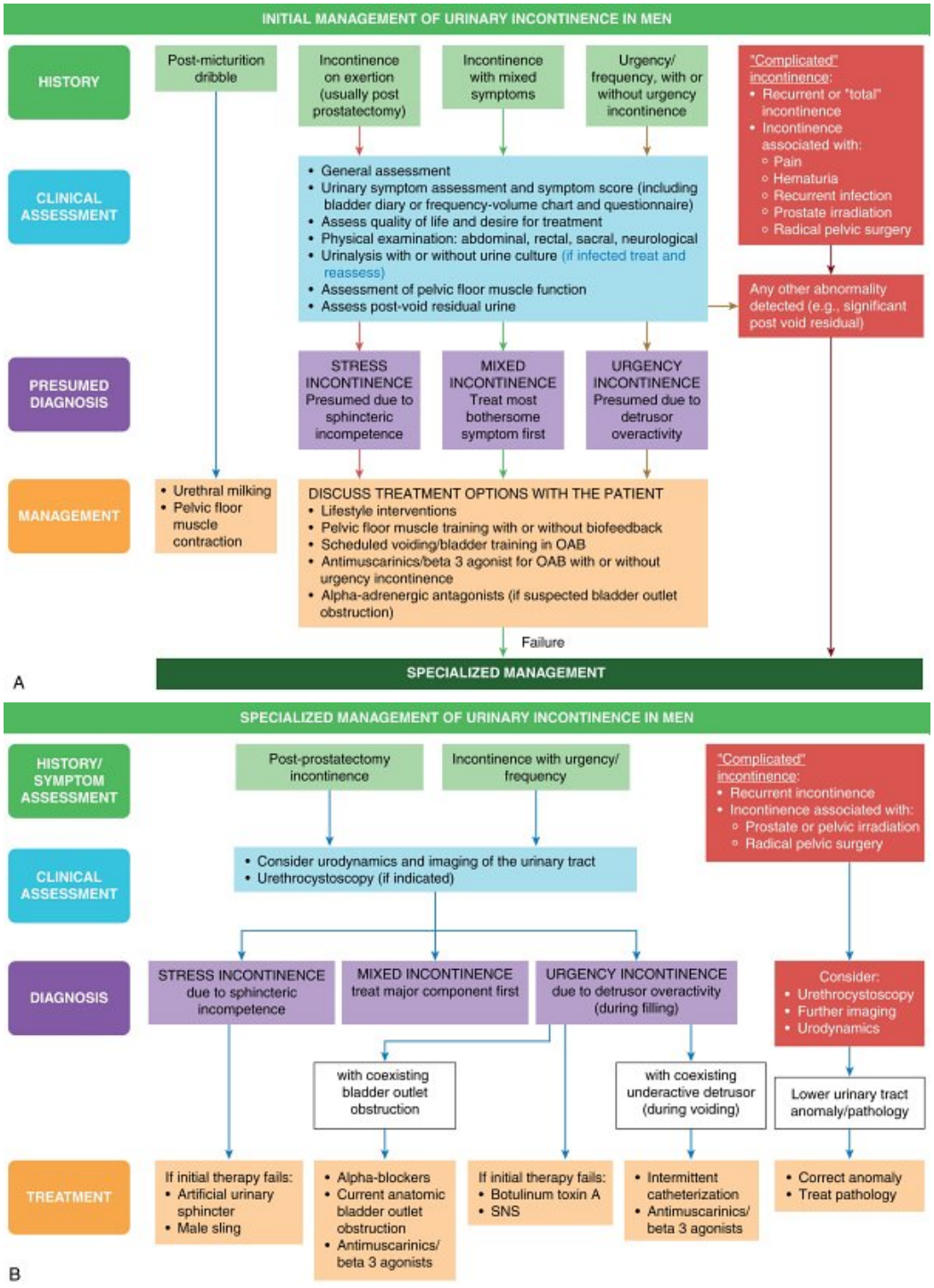


FIG. 113.1

Chapter Review

1. UI is defined by the International Continence Society as the complaint of any involuntary loss of urine. It is part of the storage lower urinary tract symptoms (LUTS).
2. UI can result from a variety of causes and can become a social and hygienic problem that affects quality of life (QoL).
3. Potential risk factors for UI include increased age, infection, functional and cognitive impairment, neurological disorders, diabetes, prostatectomy, drugs, and alcohol.
4. UI in men is prevalent but less so than in women, and it can reach up to 39% of all men.
5. There are different types of UI in men, including SUI, UUI, a mixture of the two, and several other types. It is important to know the differences among them, as this affects management.
6. It is important to assess these patients appropriately to formulate a management plan that will help improve the patient's QoL. Depending on the etiology, initial treatment is usually conservative and medical, with surgical therapies reserved for those who fail initial treatment and in whom the UI is affecting their QoL.

114

Urodynamic and Video-Urodynamic Evaluation of the Lower Urinary Tract

Benjamin M. Brucker, and Victor W. Nitti

Keywords

bladder outlet obstruction; bladder storage; CMG; compliance; cystometrogram; detrusor overactivity; emptying; incontinence; leak point pressure; neurogenic bladder; overactive bladder; PVR; stress incontinence; UDS; urodynamics; uroflow; video-urodynamics

Questions

1. Indications for urodynamic studies:
    a. are supported by high-quality, level 1 evidence for most conditions.
    b. are better defined for men than for women.
    c. are best defined by the clinician who has clear-cut reasons for performing the study and will use the information obtained to guide treatment.
    d. are of little value in assessing a patient with neurogenic lower urinary tract dysfunction.
    e. are only useful in women when incontinence is seen clinically.
2. The American Urological Association/Society for Urodynamics and Female Urology Urodynamics (AUA/SUFU) guidelines and

recommendations:
   a. are all supported by level 1 or 2 evidence.
   b. do not apply to patients with neurogenic lower urinary tract dysfunction.
   c. are intended to assist the clinician in the appropriate selection of urodynamic tests, following evaluation and symptom characterization.
   d. include recommendations on standardization of urodynamic equipment.
   e. do not consider expert opinion.

3. Which of the following tests assesses bladder compliance?
   a. Cystometrogram (CMG)
   b. Micturitional urethral pressure profile
   c. Postvoid residual volume
   d. Voiding pressure flow study
   e. Electromyogram

4. Detrusor pressure:
   a. can be measured directly via a transurethral catheter.
   b. should remain low (near zero) during bladder filling.
   c. rises abruptly and does not return to baseline with detrusor overactivity.
   d. rises before the external sphincter relaxes in normal voluntary micturition.
   e. is obtained by adding the abdominal pressure to the vesicle pressure.

5. Detrusor overactivity (DO):
   a. is synonymous with overactive bladder.
   b. is necessary to make a diagnosis of urodynamic bladder outlet obstruction.
   c. can be seen on urodynamics of asymptomatic men and women.
   d. is commonly associated with renal deterioration.
   e. is a diagnosis made based on history alone.

6. The hallmark of bladder outlet obstruction is:
   a. incomplete bladder emptying.
   b. low pressure–low flow voiding dynamics.

c. high pressure–low flow voiding dynamics.
d. impaired detrusor contractility.
e. elevated postvoid residual.

7. The external urethral sphincter should normally:
   a. relax with an involuntary bladder contraction in a neurologically normal person.
   b. relax prior to a voluntary detrusor contraction in a neurologically normal person.
   c. progressively relax while the bladder fills.
   d. always contract when the detrusor contracts.
   e. contract during urination in a neurologically normal person.
8. Video-urodynamics (VUDS):
   a. is the most precise measure of lower urinary tract function and should be used in all cases in which urodynamics are to be performed.
   b. is required to assess obstruction in a man.
   c. is the procedure of choice for documenting bladder neck dysfunction in men and women.
   d. is impractical to perform in spinal cord–injured patients.
   e. is needed to confirm detrusor overactivity.
9. Urethral function tests such as abdominal leak point pressure (ALPP) and maximum urethral closure pressure (MUCP):
   a. can precisely define intrinsic sphincter deficiency (ISD).
   b. should be done routinely before all surgery for stress incontinence.
   c. are the most important part of the urodynamic assessment of women with stress urinary incontinence (SUI).
   d. should not be used as a single factor to grade the severity of incontinence.
   e. can only be performed during ambulatory urodynamics.
10. Which of the following conditions/factors may result in inaccurate measurement of bladder compliance?
   a. Prior radiation to the pelvis
   b. Use of fluid-filled urodynamic catheters
   c. Presence of vesicoureteral reflux
   d. Bladder outlet obstruction

e. History of genitourinary (GU) tuberculosis (TB)

11. According to the Functional Classification System, the symptom of stress incontinence can be classified as:
    a. failure to store secondary to an overactive bladder outlet.
    b. failure to store secondary to an overactive bladder.
    c. failure to store secondary to an underactive bladder outlet.
    d. failure to empty secondary to an underactive bladder outlet.
    e. failure to empty secondary to an overactive bladder outlet.
12. During multichannel urodynamics, what is the best measure that allows the clinician to look for abdominal straining occurring during micturition?
    a. Rectal or vaginal catheter pressure
    b. Bladder catheter pressure
    c. Uroflow pattern
    d. Electromyogram activity and uroflow velocity
    e. Postvoid residual
13. For women with stress incontinence, urodynamics have the most useful role in which of the following scenarios?
    a. In women who are considering surgical correction who also have urgency incontinence symptoms or difficulty emptying the bladder
    b. In predicting outcomes of surgery for women with pure stress incontinence
    c. In predicting the likelihood of voiding dysfunction in women with pure stress incontinence
    d. In predicting outcomes of conservative, nonsurgical treatments for women with mixed incontinence
    e. Prior to starting conservative, nonsurgical treatments for women with pure incontinence

Answers

1. **c. Are best defined by the clinician who has clear-cut reasons for performing the study and will use the information obtained to guide treatment.** Urodynamics studies have been used for decades,

yet clear-cut, level-1 evidenced-based “indications” for its use are surprisingly lacking. There are a number of reasons for this. It is difficult to conduct proper randomized controlled trials on urodynamics for conditions in which lesser levels of evidence and expert opinion strongly suggest clinical utility and in which “empiric treatment” is potentially harmful or even life-threatening (e.g., neurogenic voiding dysfunction). In addition, symptoms can be caused by a number of different conditions, and it is difficult to study pure or homogeneous patient populations. Given the current state of evidence for urodynamic studies, what is most important is that the clinician has clear-cut reasons for performing the study and that the information obtained will be used to guide treatment of the patient. Despite having established nomograms for bladder outlet obstruction in men, the indications for urodynamics in men are no more clear-cut than they are in women. Urodynamics probably have the most important role in the diagnosis and management of patients with neuropathic voiding dysfunction. In straightforward cases of pure SUI where incontinence is seen clinically, urodynamics may not be needed.

2. **c. Are intended to assist the clinician in the appropriate selection of urodynamic tests, following evaluation and symptom characterization.** The intent of the guideline was to assist clinicians in the appropriate selection of tests rather than to provide absolute indications for urodynamics. The review of literature produced few articles offering a high level of evidence, and most of the recommendations are based on lower levels of evidence, expert opinion, and clinical principles.
3. **a. Cystometrogram (CMG).** Compliance is the change in bladder volume/change in bladder pressure. Bladder pressure during filling is assessed by CMG.
4. **b. Should remain low (near zero) during bladder filling.** Detrusor pressure normally remains low during filling, as the bladder is highly compliant. It cannot be measured directly with a transurethral catheter, but must be obtained via subtraction of abdominal pressure from vesicle pressure. With detrusor

overactivity, pressure usually returns to baseline after the involuntary contraction abates.

5. **c. Can be seen on urodynamics of asymptomatic men and women.** DO is defined as involuntary bladder contractions on CMG. Overactive bladder is a symptom complex with the hallmark symptom of urinary urgency and is not diagnosed on urodynamics. DO has been reported to occur in studies on asymptomatic men and women. Impaired compliance, not DO, is associated with renal deterioration. DO is an observation made during urodynamics.
6. **c. High pressure–low flow voiding dynamics.** Obstruction is defined by high pressure low flow voiding. It may or may not be accompanied by incomplete bladder emptying. Impaired detrusor contractility may sometimes, but not always, be a long-term consequence of obstruction.
7. **b. Relax prior to a voluntary detrusor contraction in a neurologically normal person.** In a neurologically normal person, the external sphincter progressively contracts with bladder filling and will also contract during an involuntary bladder contraction (guarding reflex). External sphincter relaxation is the first step in the micturition cycle and precedes the detrusor contraction. In detrusor external sphincter dyssynergia, an abnormal neurologic condition, the external sphincter contracts when the detrusor does. Intermittent or fluctuating flow rate of urine due to intermittent contractions of the external sphincter in a neurologically normal person is considered dysfunctional voiding.
8. **c. Is the procedure of choice for documenting bladder neck dysfunction in men and women.** Although VUDS provides the most precise evaluation of voiding function and dysfunction and is particularly useful when anatomic structure and function are important, it is not practical or necessary for all centers to have VUDS capabilities. VUDS is useful for a number of conditions when an accurate diagnosis cannot otherwise be obtained (e.g., by conventional urodynamics), including complicated voiding dysfunction or known or suspected neuropathic voiding dysfunction (adults and children), unexplained urinary retention in women, prior radical pelvic surgery, urinary diversion, preceding or

following renal transplant, or prior pelvic radiation. VUDS is the procedure of choice for documenting bladder neck dysfunction in men and women. VUDS is not specifically needed to diagnose DO.

9. **d. Should not be used as a single factor to grade the severity of incontinence.** Urethral function tests such as ALPP and MUCP have not been shown to be consistently useful in defining ISD or outcomes of treatments for SUI. They may be useful for some clinicians, but are by no means mandatory. According to the International Continence Society, "Urethral function measurements of leak point pressures and urethral closure pressures are not used as a single factor to grade the severity of incontinence." Urodynamics are not necessary before surgical treatment of SUI for all women, but if it is done, the AUA/SUFU Guideline states that ALPP or MUCP should be preformed. ALPP can be reported as part of ambulatory urodynamics; however, it is not a measure that is unique to this method.
10. **c. Presence of vesicoureteral reflux.** Bladder outlet obstruction and radiation can cause a decrease in compliance, but should not affect its accurate measurement. GU TB also can cause significant bladder fibrosis and impaired compliance. Reflux can make compliance look worse than it actually is secondary to the "pop-off" it creates. Fluoroscopy or VUDS is necessary in some cases (e.g., neurogenic bladder) to assess for reflux during filling. In addition, filling rate and involuntary detrusor contractions can also make compliance look worse than it actually is. There are no data to support the proposition that fluid-filled catheters or air charge catheters will result in inaccurate measures of compliance, which is a measure of change volume over pressure.
11. **c. Failure to store secondary to an underactive bladder outlet.** Stress incontinence is a symptom caused by failure to store urine during increases in abdominal pressure. It can be caused by a loss of outlet resistance or an underactive bladder outlet.
12. **a. Rectal or vaginal catheter pressure.** Rectal and or vaginal pressure are used to measure abdominal pressure throughout urodynamics. Patients who are straining to void will exhibit increases in abdominal pressure measurements. Flow patterns can

be suggestive of abdominal straining, but are not as accurate as measuring abdominal pressure. Postvoid residual may allow clinicians to understand how well the patient is emptying his or her bladder but does not give information about how voiding is accomplished. Similarly, EMG may be increased from abdominal straining, but there are many other causes of increased EMG activity during voiding. Bladder catheter pressure will increase with abdominal straining. The difference between the bladder catheter and the rectal or vaginal catheter allows for calculation of the detrusor pressure.

13. **a. In women who are considering surgical correction who also have urgency incontinence symptoms or difficulty emptying the bladder.** UDS in women with pure stress or stress-predominant mixed incontinence and normal emptying has not been shown in randomized controlled trials to be more beneficial than office evaluation alone and has not been shown to predict outcomes of surgery in the same population. Women with significant mixed incontinence and emptying problems have not been studied in randomized controlled trials, and it is felt that UDS is beneficial in these women.

Chapter Review

1. Urodynamics are performed in an unnatural setting and therefore does not always predict the findings with normal activity.
2. Normal uroflow is a bell-shaped curve.
3. EMG patch electrodes measure perineal muscle function with the assumption that it is reflective of urethral external sphincter function.
4. To specifically measure external sphincter function, needle electrodes must be used.
5. Mean values for compliance are 40–120 mL/cm H_2O.
6. Measurement of compliance is difficult to interpret; therefore, pressures during filling are more often used to predict outcome.

7. There are two types of leak-point pressures: (a) abdominal leak-point pressure, which is defined as the intravesical pressure at which urine leakage occurs due to increased abdominal pressure; and (b) detrusor leak-point pressure, which is a measure of detrusor pressure at which urine leakage occurs in the absence of a detrusor contraction or increased abdominal pressure. This measure is generally used in patients with decreased compliance or lower motor neuron disease.
8. Detrusor pressures that are sustained above 40 cm H_2O lead to deterioration of the upper tracts.
9. Maximum urethra closure pressure is defined as the difference between peak urethral pressure and intravesical pressure and is normally between 40 and 60 cm H_2O.
10. Bladder outlet obstruction index is defined by the equation: BOOI = Pdet Qmax − 2(Qmax). In men, a value greater than 40 is considered obstructed; a value less than 20 is considered unobstructed.
11. A uroflow less than 12 mL/sec and Pdet greater than 25 cm H_2O predict outlet obstruction in women.
12. Detrusor external sphincter dyssynergia can be due to a neurologic lesion (above the sacral micturition center) or a learned disorder. The latter is considered dysfunctional voiding.
13. Internal sphincter dyssynergia must be diagnosed by VUDS.
14. Stress incontinence, which is observed only when a coexisting prolapse is reduced, is referred to as occult or latent stress incontinence.
15. For internal sphincter dyssynergia to occur, the spinal cord lesion must be above the sympathetic outflow (T-10-L-1).
16. VUDS is the procedure of choice for documenting bladder neck dysfunction in men and women.
17. In a neurologically normal person, the external sphincter progressively contracts with bladder filling and will also contract during an involuntary bladder contraction (guarding reflex). External sphincter relaxation is the first step in the micturition cycle and precedes the detrusor contraction.

115

Urinary Incontinence and Pelvic Prolapse

Epidemiology and Pathophysiology

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Keywords

epidemiology; pathophysiology; pelvic organ prolapse; stress urinary incontinence; urgency urinary incontinence

Questions

1. An example of a lower urinary tract (LUT) symptom is:
    a. urine loss while playing tennis.
    b. urodynamic stress incontinence with Valsalva leak point pressure (VLPP) of 60 cm H_2O.
    c. a symptom score of 19 on the AUA Symptom Index.
    d. the finding of urinary incontinence on a supine stress test.
    e. the presence of stage 2 anterior pelvic organ prolapse on a cystogram.
2. Which of the following is consistent with the diagnosis of urgency urinary incontinence?
    a. Leakage of urine with coughing and a VLPP of 60 cm H_2O
    b. Leakage of urine with detrusor overactivity

c. Leakage of urine while coughing and detrusor leak point pressure of 50 cm H_2O

d. Leakage of urine while coughing and leakage of urine with urgency

e. Leakage of urine and feces while straining

3. The most common form of urinary incontinence (UI) in a woman aged 40 years is:
   a. stress incontinence.
   b. urgency incontinence.
   c. mixed incontinence.
   d. detrusor overactivity incontinence.
   e. continuous incontinence.
4. What factor has been associated with LUTS but not urinary incontinence in women?
   a. Advancing age.
   b. Caucasian race.
   c. Number of prior vaginal delivereries.
   d. Caffeine intake.
   e. Large birth weight.
5. Later in life, when compared with a woman who had her first child at the age of 36, a woman delivering her first child at 23 years of age would be:
   a. more likely to experience SUI.
   b. equally likely to experience SUI.
   c. more likely to experience UUI.
   d. equally likely to experience UUI.
   e. less likely to experience MUI.
6. The symptom most closely associated with the presence of advanced pelvic organ prolapse is the sensation of:
   a. pelvic pressure.
   b. pelvic pain.
   c. constipation.
   d. voiding difficulty.
   e. vaginal bulge.

7. Which of the following Pelvic Organ Prolapse Quantification System (POP-Q) scores is implausible?
    a. Aa of −2
    b. Ap of +4
    c. Ba of +5
    d. Bp of 0
    e. C of −8
8. A woman with POP-Q scores of Aa −2, Ba 0, C −5, Ap +1, Bp +1 would be considered to have what stage prolapse?
    a. Stage 0
    b. Stage 1
    c. Stage 2
    d. Stage 3
    e. Stage 4
9. A patient with a POP-Q score of Aa −2 , Ba −2, Ap +2, Bp +2, and C 0 should be counseled to undergo:
    a. anterior colporrhaphy.
    b. posterior colporrhaphy.
    c. anterior and posterior colporrhaphy.
    d. posterior colporrhaphy and sacrospinous fixation.
    e. sacrospinous fixation.
10. During bladder contraction, which is most active?
    a. Pelvic nerve
    b. Hypogastric nerve
    c. Pudendal nerve
    d. A-delta nerves
    e. Onuf nucleus

Answers

1. A LUT symptom is one that is recognizable by the patient. A sign is one that is observable by a clinician (i.e., urine loss during a supine stress test). A VLPP result is a urodynamic observation, whereas a cystogram report of a cystocele is a radiologic observation.

2. **b. Leakage of urine with detrusor overactivity.** Urgency urinary incontinence may be associated with detrusor overactivity (DO), though DO need not be present for the diagnosis, merely sense of urgency with loss of urine. Mixed incontinence is the symptomatic complaint of both stress urinary incontinence and urgency incontinence.
3. **a. Stress incontinence.** Stress urinary incontinence (SUI) is the most common form of UI in young women. In contrast, urge UI and mixed UI appear to become more prevalent forms of UI with aging. Continuous incontinence is uncommon—often associated with iatrogenic injuries to the lower urinary tract resulting in fistulae.
4. **d. Caffeine intake.** Caffeine consumption has been associated with urinary urgency, but most studies have not necessarily noted an association with urinary incontinence. All other factors mentioned have been associated with a form of UI.
5. **a. More likely to experience SUI.** Younger age at time of first pregnancy has been associated with a greater likelihood of developing SUI later in life.
6. **e. Vaginal bulge.** Although all of the symptoms described can be associated with POP, only the specific complaint of a vaginal bulge has been consistently demonstrated to be associated with the presence of POP. All others can be associated with other conditions distinct from the presence of POP. For example, pelvic pressure can be related to pelvic floor dysfunction or a variety of other conditions, and pelvic pain can be associated with various forms of pelvic pathology including adnexal pathology.
7. **b. Ap of + 4.** POP-Q points are measured by their distance from the hymenal ring (in centimeters). By definition, Aa and Ap are the points 3 cm from the hymen on the anterior and posterior vaginal walls, respectively. It would be impossible for this point to be greater than 3 cm proximal (or distal) to the hymen. In advanced prolapse, this point could be up to 3 cm distal to the hymen. C point represents the distance from the hymen to the cervix or vaginal cuff (post hysterectomy). Ba and Bp point represent the leading edge of the most advanced aspect of the prolapse on the anterior and

posterior vaginal walls, respectively. As such, they can be well beyond + 3 in advanced prolapse.

8. **c. Stage 2.** Staging in POP-Q is based on the leading edge of prolapse. This equates to the highest (positive) number associated with the points listed (Aa, Ba, Ap, Bp, C). The leading edge of the most advanced POP is greater than 1 cm proximal to the hymen in patients with stage 1 prolapse. Stage 2 patients will have their leading edge between 1 cm proximal to and distal to the hymen. Stage 3 indicates a leading edge greater than 1 cm beyond the hymen but not completely everted (stage 4).
9. **d. Posterior colporrhaphy and sacrospinous fixation.** This patient has both posterior compartment prolapse (based on Ap and Bp points beyond the hymen) and apical prolapse (C point at the hymen) based on the POP-Q score. Therefore the patient should be counseled to have both posterior repair of some type (posterior colporrhaphy in this example) and apical repair (sacrospinous fixation in this example).
10. **a. Pelvic nerve.** During bladder contraction, parasympathetic transmission (via the pelvic nerve) is active and sympathetic transmission (via the hypogastric nerve) is suppressed. Pudendal innervation to the external sphincter is suppressed. Pudendal innervation is derived from Onuf nucleus in the sacral cord. A-delta nerves (afferent) are active during storage and are involved in the spinal reflex mechanism that promotes closure of the bladder neck.

Chapter Review

1. Pelvic organ prolapse may mask incontinence.
2. The prevalence of urinary incontinence in women is between 25% and 40%. Ten percent of women experience weekly incontinence episodes. Fecal incontinence occurs in 17% of women with pelvic organ prolapse.
3. The female bladder neck is weaker than the male bladder neck and is often incompetent.

4. In women, the majority of the urethra should be considered an active area of sphincter control; however, in the female the most important portion of continence is the mid-urethra.
5. A Valsalva leak point pressure of less than 60 cm H_2O indicates but does not confirm intrinsic sphincter dysfunction.
6. Twelve percent of men report terminal dribbling.
7. Rates of overactive bladder increase with age.
8. Oral estrogen treatment is associated with the development of incontinence. Topical estrogen therapy is not linked to stress incontinence.
9. Intrinsic properties of the urethra mucosa and urethra wall are important in maintaining continence in women.
10. The anterior vaginal wall provides posterior support of the urethra allowing for compression of the mid-urethra. The urethra is attached laterally to the arcus tendineus by the urethra pelvic ligaments.
11. Urge UI and mixed UI appear to become more prevalent forms of UI with aging.
12. Age is strongly correlated with the development of UI of all types. White women appear to be at greater risk for UI when compared with African American and Asian women in particular.
13. POP-Q points are measured by their distance from the hymenal ring (in centimeters). By definition, Aa and Ap are the points 3 cm from the hymen on the anterior and posterior vaginal walls, respectively. Ba and Bp point represent the leading edge of the most advanced aspect of the prolapse on the anterior and posterior vaginal walls, respectively. Thus they can be well beyond + 3 in advanced prolapse. C point represents the distance from the hymen to the cervix or vaginal cuff (post hysterectomy).
14. The leading edge of the most advanced POP is greater than 1 cm proximal to the hymen in patients with stage 1 prolapse. Stage 2 patients will have their leading edge between 1 cm proximal to and distal to the hymen. Stage 3 indicates a leading edge greater than 1 cm beyond the hymen but not completely everted.

116

Neuromuscular Dysfunction of the Lower Urinary Tract

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Keywords

cerebrovascular accident; myelomeningocele; neurogenic bladder; neuroplasticity; spinal cord injury; urodynamics; voiding dysfunction

Questions

1. What is the general pattern of voiding dysfunction secondary to neurologic lesions above the level of the brainstem?
    a. Involuntary bladder contractions, smooth sphincter dyssynergia, striated sphincter synergy
    b. Involuntary bladder contractions, smooth sphincter synergy, striated sphincter synergy
    c. Involuntary bladder contractions, smooth sphincter synergy, striated sphincter dyssynergia
    d. Detrusor hypocontractility, smooth sphincter synergy, striated sphincter synergy
    e. Detrusor areflexia, smooth sphincter synergy, striated sphincter synergy
2. What is the general pattern of voiding dysfunction that results from complete lesions of the spinal cord above the level of S2 after recovery from spinal shock?

a. Involuntary bladder contractions, smooth sphincter dyssynergia, striated sphincter synergy
b. Involuntary bladder contractions, smooth sphincter synergy, striated sphincter synergy
c. Involuntary bladder contractions, smooth sphincter synergy, striated sphincter dyssynergia
d. Detrusor hypocontractility, smooth sphincter synergy, striated sphincter synergy
e. Detrusor areflexia, smooth sphincter synergy, striated sphincter synergy

3. Which of the following is the most common long-term expression of lower urinary tract dysfunction after a cerebrovascular accident (CVA)?
   a. Detrusor areflexia
   b. Lack of sensation of filling
   c. Impaired bladder contractility
   d. Striated sphincter dyssynergia
   e. Detrusor overactivity
4. A 65-year-old man who has sustained a stroke but is otherwise in good health has symptoms of hesitancy, straining to void, urgency, and frequency. The optimal next step in management is:
   a. anticholinergic therapy.
   b. transurethral resection of the prostate (TURP).
   c. transurethral incision of the bladder neck and prostate.
   d. clean intermittent catheterization.
   e. full urodynamic evaluation.
5. The most common pattern of micturition in children and adults who have cerebral palsy (CP) and no other complicating neurologic condition is:
   a. abnormal filling/storage because of detrusor overactivity; normal emptying.
   b. normal filling/storage; normal emptying.
   c. normal filling/storage; abnormal emptying because of smooth sphincter dyssynergia.
   d. normal filling/storage; abnormal emptying because of striated sphincter dyssynergia.

e. abnormal filling/storage because of detrusor overactivity; abnormal emptying because of striated sphincter dyssynergia.

6. The most common urodynamic findings in individuals with CP who do exhibit lower urinary tract dysfunction are:
   a. detrusor areflexia, coordinated sphincters.
   b. detrusor overactivity, smooth sphincter dyssynergia, striated sphincter dyssynergia.
   c. detrusor overactivity, smooth sphincter synergy, striated sphincter dyssynergia.
   d. decreased detrusor compliance, coordinated sphincters.
   e. detrusor overactivity, coordinated sphincters.
7. The most common urodynamic abnormality found in patients with voiding dysfunction secondary to Parkinson disease (PD) is:
   a. impaired sensation during filling.
   b. striated sphincter dyssynergia.
   c. striated sphincter bradykinesia.
   d. detrusor overactivity.
   e. impaired detrusor contractility.
8. Urodynamic findings that may distinguish multiple system atrophy (MSA) from PD include:
   a. sphincteric insufficiency
   b. detrusor overactivity
   c. detrusor areflexia
   d. lack of sensation during filling
   e. decreased bladder capacity
9. The most common urodynamic findings in individuals with multiple sclerosis (MS) who have lower urinary tract dysfunction are:
   a. detrusor overactivity, smooth sphincter dyssynergia
   b. detrusor overactivity, smooth sphincter synergia
   c. detrusor areflexia, impaired filling sensation
   d. detrusor areflexia, striated sphincter dyssynergia
   e. normal filling and storage
10. The incidence of upper urinary tract deterioration is greatest in which of the following?
   a. MS
   b. MSA

c. PD
d. Spinal cord injury (SCI)
e. Diabetes

11. All of the following are risk factors for upper urinary tract deterioration in a patient with a suprasacral SCI except:
    a. high-pressure storage.
    b. high detrusor leak-point pressure.
    c. chronic bladder overdistention.
    d. high abdominal leak-point pressure.
    e. vesicoureteral reflux with infection.
12. In spinal shock, findings generally include all of the following EXCEPT:
    a. acontractile bladder.
    b. areflexic bladder.
    c. open bladder neck.
    d. absent guarding reflex.
    e. maximal urethral closure pressure above normal.
13. The presence of true striated sphincter dyssynergia implies a neurologic lesion between the:
    a. pons and the sacral spinal cord.
    b. cerebral cortex and the pons.
    c. cervical and the sacral spinal cord.
    d. sacral spinal cord and the striated sphincter.
    e. cauda equina and the striated sphincter.
14. Which of the following is least characteristic as a finding in autonomic hyperreflexia?
    a. Headache before bladder contraction
    b. Hypertension
    c. Flushing above the level of the lesion
    d. Tachycardia
    e. Sweating above the level of the lesion
15. A patient with a history of cervical SCI is undergoing urodynamic evaluation and currently has 350 cc infused volume. He begins to complain of headache and develops hypertension and bradycardia. What is the next step?
    a. Continue filling and reassure him

b. Administer an antibiotic
c. Give terazosin 5 mg immediately
d. Empty his bladder
e. Apply 2% nitroglycerin paste

16. The sacral spinal cord terminates in the cauda equina at approximately the spinal column level of:
    a. T10.
    b. L1.
    c. L2.
    d. L3.
    e. S1.
17. Patients who have voiding dysfunction secondary to lumbar disk disease most commonly present with which of the following symptoms and urodynamic findings?
    a. Retention; involuntary bladder contractions
    b. Incontinence; involuntary bladder contractions
    c. Retention; decreased bladder compliance
    d. Difficulty voiding; normal bladder compliance
    e. Incontinence; normal bladder compliance
18. The clinical picture of cauda equina syndrome (CES) varies widely. Which of the following is typically not seen with CES?
    a. Urinary incontinence
    b. Loss of bowel control
    c. Perineal sensory loss
    d. Sexual dysfunction
    e. Limb spasticity
19. The combination that best describes the type of permanent voiding dysfunction that can occur after radical pelvic surgery is:
    a. stress incontinence; detrusor areflexia.
    b. urgency incontinence; detrusor overactivity.
    c. reflex incontinence; detrusor areflexia.
    d. urgency incontinence; detrusor overactivity.
    e. stress incontinence; detrusor overactivity.
20. What is optimal management for a 65-year-old man who has had a first occurrence of urinary retention after radical pelvic surgery?
    a. Anticholinergic therapy

b. Clean intermittent catheterization
c. TURP
d. External sphincterotomy
e. Bethanechol chloride

21. The urodynamic parameter most likely to distinguish urinary retention due to prostatic obstruction from urinary retention due to "classic diabetic cystopathy" is:
    a. uroflow.
    b. residual urine volume.
    c. bladder compliance.
    d. vesical pressure.
    e. detrusor pressure.
22. Detrusor-striated sphincter dyssynergia is least expected to occur with which of the following conditions?
    a. MS
    b. SCI
    c. Stroke
    d. Autonomic hyperreflexia
    e. Transverse myelitis
23. Treatment options for DSD can include all of the following except:
    a. intermittent catheterization
    b. injection of onabotulinum toxin into the sphincter
    c. sacral neuromodulation
    d. sphincterotomy
    e. urinary diversion
24. Differentiation of bladder neck obstruction from dysfunctional voiding is most easily and accurately made by:
    a. filling cystometry.
    b. voiding cystometry.
    c. cystourethroscopy.
    d. flowmetry and residual urine determination.
    e. video-urodynamics study.
25. According to the AUA consensus statement on non-neurogenic urinary retention,
    a. non-neurogenic urinary retention is defined as a postvoid residual > 300 mL on a single occasion.

b. low-risk and high-risk patients should be distinguished and patients with high-risk features should be treated.
c. low-risk patients include those with hydronephrosis, recurrent UTIs, and chronic kidney disease.
d. if long-term catheterization is needed, then an indwelling urethral catheter is preferred.
e. patients with low-risk urinary retention do not need to be followed long term.

26. Which of the following is not a typical finding in a patient with Fowler syndrome?
    a. Female younger than 30 years
    b. Unable to void for a day or more with no urgency
    c. Bladder capacity of less than 1 L
    d. Increasing lower abdominal discomfort
    e. Electromyographic (EMG) abnormalities
27. Which of the following with the typical history is the most specific study to make the diagnosis of Fowler syndrome?
    a. Striated sphincter needle EMG recording
    b. Striated sphincter patch EMG recording
    c. Neurologic examination
    d. Spinal magnetic resonance imaging (MRI) examination
    e. Detrusor pressure/urinary flow recording
28. Which of the following has proved to be successful in treating the urologic manifestations of Fowler syndrome?
    a. Estrogen therapy
    b. Progesterone therapy
    c. Baclofen therapy
    d. Botulinum toxin injection therapy
    e. Neuromodulation
29. A 65-year-old female presents with a history of pelvic radiation for treatment of cervical cancer 30 years ago. What urodynamic findings would you expect?
    a. Reduced bladder capacity
    b. Increased postvoid residual
    c. Normal compliance
    d. Decreased flow rate

e. Bladder outlet obstruction

30. In the medical management of neurogenic detrusor overactivity (NDO), which of the following is least likely to worsen constipation?
    a. Oxybutynin
    b. Mirabegron
    c. Tolterodine
    d. Tamsulosin
    e. Trospium
31. Increasing numbers of individuals with myelomeningocele are surviving into adulthood and older age. Critical for stability of function is recognition that the spinal cord may be tethered. Regarding the tethered spinal cord,
    a. the level of tethering as demonstrated by MRI is predictive of findings on urodynamics.
    b. urodynamics often shows improvement after detethering.
    c. tethered cord syndrome is not usually associated with bowel or leg dysfunction.
    d. there are no predictors for social continence after detethering.
    e. the symptomatic tethered cord, associated with worsening ambulation, can be managed non-operatively.
32. Guillain-Barré syndrome is a demyelinating disorder. It is:
    a. not usually identified as a clinical entity having sporadic clinical manifestation.
    b. not caused by an acute inflammatory disorder.
    c. not usually associated with an antecedent acute infectious illness.
    d. associated with lower urinary tract function, usually in less than 5% of cases.
    e. associated with voiding dysfunction, which is usually reversible
33. In a patient with voiding dysfunction secondary to myelomeningocele who has slightly decreased compliance, detrusor areflexia, and low-pressure moderate to severe vesicoureteral reflux, which of the following urodynamic changes would be most likely after ureteral reimplantation alone?

a. Compliance increased
b. Voiding pressure decreased
c. Valsalva leak-point pressure decreased
d. Maximum urethral closure pressure decreased
e. Maximum bladder capacity decreased

34. Which of the following is true about patients with myelomeningocele?
    a. Myelomeningoceles most commonly occur at the cervical level.
    b. The level of the lesion correlates poorly with urodynamic findings.
    c. Neurologic examination of patients with myelomeningocele accurately predicts urodynamic findings.
    d. Patients rarely have lower urinary tract dysfunction.
    e. Life-long follow-up is not necessary.
35. Findings consistent with herpes virus infection may include all of the following except:
    a. vesicles in the bladder on cystoscopy
    b. urinary retention
    c. detrusor overactivity
    d. rash on the palms and soles of feet
    e. fever, malaise

Answers

1. **b. Involuntary bladder contractions, smooth sphincter synergy, striated sphincter synergy.** Neurologic lesions above the level of the brainstem that affect micturition generally result in involuntary bladder contractions with smooth and striated sphincter synergy. Sensation and voluntary striated sphincter function are generally preserved. Areflexia may occur either initially or as a permanent dysfunction.
2. **c. Involuntary bladder contractions, smooth sphincter synergy, striated sphincter dyssynergia.** Patients with complete lesions of the spinal cord between spinal cord levels T6 and S2, after they

recover from spinal shock, generally exhibit involuntary bladder contractions without sensation of the contraction, smooth sphincter synergy, but striated sphincter dyssynergia. Those with lesions above T6 may experience, in addition, smooth sphincter dyssynergia and autonomic hyperreflexia.

3. **e. Detrusor overactivity.** The most common long-term expression of lower urinary tract dysfunction after a CVA is detrusor overactivity. Sensation is variable but is classically described as generally intact, and thus the patient has urgency and frequency with detrusor overactivity.
4. **e. Full urodynamic evaluation.** Poor flow rates and high residual urine volumes in a male with pre-CVA symptoms of prostatism generally indicate prostatic obstruction, but a full urodynamic evaluation is advisable before committing a patient to mechanical outlet reduction primarily to exclude detrusor hyperactivity with impaired contractility as a cause of symptoms.
5. **b. Normal filling/storage; normal emptying.** Most children and adults with only CP have urinary control and what seems to be normal filling/storage and normal emptying. The actual incidence of voiding dysfunction is somewhat vague because the few available series report findings predominantly in those who present with voiding symptoms. One study estimated that a third or more of children with CP are so affected. When an adult with CP presents with an acute or subacute change in voiding status, however, it is most likely unrelated to CP.
6. **e. Detrusor overactivity, coordinated sphincters.** In those individuals with CP who exhibit significant dysfunction, the type of damage that one would suspect from the most common urodynamic abnormalities seems to be localized above the brainstem. This is commonly reflected by detrusor overactivity and coordinated sphincters. Spinal cord damage can occur, however, and probably accounts for those individuals with CP who seem to have evidence of striated sphincter dyssynergia.
7. **d. Detrusor overactivity.** The most common urodynamic finding is detrusor overactivity. The pathophysiology of detrusor overactivity most widely proposed is that the basal ganglia normally have an

inhibitory effect on the micturition reflex, which is abolished by the cell loss in the substantia nigra.

8. **a. Sphincteric insufficiency.** Patients with MSA usually have sphincteric insufficiency, while patients with PD rarely have external sphincter acontractility. For this reason, a bladder outlet treatment is rarely indicated in MSA patients. Decreased compliance may also occur in MSA.
9. **b. Detrusor overactivity and smooth sphincter synergia.** In terms of urodynamic findings, detrusor overactivity is the most common abnormality detected and, in general, the smooth sphincter is synergic and sensation is intact. There can also be coexisting striated sphincter dyssynergia and/or impaired detrusor contractility.
10. **d. Spinal cord injury (SCI).** Progressive neurologic diseases cause upper tract damage much less commonly than SCI, even when associated with severe disability and spasticity.
11. **d. High abdominal leak-point pressure.** As with all patients with neurologic impairment, a careful initial evaluation and periodic follow-up evaluation must be performed to identify and correct the following risk factors and potential complications: bladder overdistention, high-pressure storage, high detrusor leak-point pressure, vesicoureteral reflux, stone formation (lower and upper tracts), and complicating infection, especially in association with reflux.
12. **c. Open bladder neck.** Spinal shock includes a suppression of autonomic activity and somatic activity, and the bladder is acontractile and areflexic. Radiologically, the bladder has a smooth contour with no evidence of trabeculation. The bladder neck is generally closed and competent unless there has been prior surgery or, in some cases, thoracolumbar and presumably sympathetic injury. The smooth sphincter mechanism seems to be functional. Some EMG activity may be recorded from the striated sphincter, and the maximum urethral closure pressure is lower than normal but still maintained at the level of the external sphincter zone; however, the normal guarding reflex is absent and there is no voluntary control.

13. **a. Pons and the sacral spinal cord.** A diagnosis of striated sphincter dyssynergia implies a neurologic lesion that interrupts the neural axis between the pontine-mesencephalic reticular formation and the sacral spinal cord.
14. **d. Tachycardia.** Symptomatically, autonomic hyperreflexia is a syndrome of exaggerated sympathetic activity in response to stimuli below the level of the lesion. The symptoms are pounding headache, hypertension, and flushing of the face and body above the level of the lesion with sweating. Bradycardia is a usual accompaniment, and an arrhythmia may be present.
15. **d. Empty his bladder.** Immediate treatment of autonomic dysreflexia is to remove the cause, which in this case is bladder distension. The current recommendation for the acute pharmacologic treatment of autonomic dysreflexia is placement of 2% nitroglycerin paste above the level of the lesion for rapid blood pressure reduction secondary to vasodilation after removal of the stimulus.
16. **c. L2.** Spinal column (bone) segments are numbered by the vertebral level, and these have a different relationship to the spinal cord segmental level at different locations. The sacral spinal cord begins at approximately spinal column level T12-L1. The spinal cord terminates in the cauda equina at approximately the spinal column level of L2.
17. **d. Difficulty voiding; normal bladder compliance.** A study reported on findings in 114 patients with lumbar disk protrusion who were prospectively studied. The authors found detrusor areflexia in 31 (27.2%) and normal detrusor activity in the remaining 83. All 31 patients with detrusor areflexia reported difficulty voiding with straining. Patients with voiding dysfunction generally present with these symptoms or in urinary retention. The most consistent urodynamic finding is that of a normally compliant areflexic bladder associated with normal innervation or findings of incomplete denervation of the perineal floor musculature.
18. **e. Limb spasticity.** CES is characterized by perineal sensory loss with loss of voluntary control of both anal and urethral sphincter and of sexual dysfunction. CES is a lower motor neuron disease, so

would not expect limb spasticity which is a characteristic of upper motor neuron disease.

19. **a. Stress incontinence; detrusor areflexia.** When permanent voiding dysfunction occurs after radical pelvic surgery, the pattern is generally one of a failure of voluntary bladder contraction, or impaired bladder contractility, with obstruction by what seems urodynamically to be residual fixed striated sphincter tone, which is not subject to voluntarily induced relaxation. Often, the smooth sphincter area is open and nonfunctional. Decreased compliance is common in these patients, and this, with the "obstruction" caused by fixed residual striated sphincter tone, results in both storage and emptying failure. These patients often experience leaking across the distal sphincter area and, in addition, are unable to empty the bladder because although intravesical pressure may be increased, there is nothing that approximates a true bladder contraction. The patient often presents with urinary incontinence that is characteristically most manifest with increases in intra-abdominal pressure. This is usually most obvious in females because the prostatic bulk in males often masks an equivalent deficit in urethral closure function. Alternatively, patients may present with variable degrees of urinary retention.
20. **b. Clean intermittent catheterization.** The temptation to perform a prostatectomy should be avoided unless a clear demonstration of outlet obstruction at this level is possible. Otherwise, prostatectomy simply decreases urethral sphincter function and thereby may result in the occurrence or worsening of sphincteric urinary incontinence. Most of these dysfunctions will be transient, and the temptation to "do something" other than perform clean intermittent catheterization initially after surgery in these patients, especially in those with little or no preexisting history of voiding dysfunction, cannot be too strongly discouraged.
21. **e. Detrusor pressure.** Detrusor contractility is classically described as being decreased in the end-stage diabetic bladder. Current evidence points to both sensory and motor neuropathy as being involved in the pathogenesis, the motor aspect per se contributing to the impaired detrusor contractility. The typically described

classic urodynamic findings include impaired bladder sensation, increased cystometric capacity, decreased bladder contractility, impaired uroflow, and, later, increased residual urine volume. The main differential diagnosis, at least in men, is generally bladder outlet obstruction because both conditions commonly produce a low flow rate. Pressure/flow urodynamic studies easily differentiate the two.

22. **c. Stroke.** True detrusor sphincter dyssynergia should exist only in patients who have an abnormality in pathways between the sacral spinal cord and the brainstem pontine micturition center, generally due to neurologic injury or disease.
23. **c. Sacral neuromodulation.** Therapy for DSD is designed to either eliminate or significantly minimize the abnormal sphincter activity or to bypass the sphincter itself.
24. **e. Video-urodynamics study.** Objective evidence of outlet obstruction in these patients is easily obtainable by urodynamic study. Once obstruction is diagnosed, it can be localized at the level of the bladder neck by video-urodynamics study, cystourethrography during a bladder contraction, or micturitional urethral profilometry.
25. **b. Low-risk and high-risk patients should be distinguished and patients with high-risk features should be treated.** Following a thorough history and physical examination, if a high-risk variable is present, such as hydronephrosis, hydroureter, chronic kidney disease, recurrent UTI, or urinary incontinence with skin sequalae, then treatment is recommended.
26. **c. Bladder capacity of less than 1 L.** The criteria for Fowler syndrome include a bladder capacity of more than 1 L with no sensation of urgency.
27. **a. Striated sphincter needle EMG recording.** *Fowler syndrome* refers particularly to a syndrome of urinary retention in young women in the absence of overt neurologic disease. The typical history is that of a woman younger than 30 years, who has found herself unable to void during the preceding day but with no sensation of urgency. MRI studies of the brain and the entire spinal cord are normal. On concentric needle electrode examination of the

striated muscle of the urethral sphincter, however, Fowler and colleagues described a unique EMG abnormality. This abnormal activity, localized to the urethral sphincter, consists of a type of activity that would be expected to cause inappropriate contraction of the muscle. Sphincter activity consists of two components: complex repetitive discharges and decelerating bursts. This abnormal activity impairs sphincter relaxation.

28. **e. Neuromodulation.** Fowler reports that efforts to treat this condition by hormonal manipulation, pharmacologic therapy, or injections of botulinum toxin have been unsuccessful. This condition is highly responsive to neuromodulation, even in women who have had retention for many months or years.
29. **a. Reduced bladder capacity.** Radiation effects on the bladder evolve over time and may be progressive. Most commonly urodynamics will demonstrate reduced bladder capacity. Patients may also have reduced compliance and detrusor overactivity.
30. **b. Mirabegron.** Mirabegron is a β3 agonist that has shown efficacy in the treatment of NDO. Mirabegron avoids the anticholinergic effects of constipation and dry mouth, but can have cardiovascular side effects including mild elevations in blood pressure and small, dose-dependent increases in heart rate.
31. **b. Urodynamics often show improvement after detethering.** Despite efforts at improved radiographic visualization of the spinal cord, imaging does not correlate with physical findings or connote overall responsiveness to surgical intervention because detethering remains a critical aspect of management and control of tethered cord. Cord tethering affects both bowel and leg function, as well as bladder function. Urodynamics is improved by detethering, and this parallels functional improvement in those individuals who have undergone the surgical procedure. The successful management of the tethered cord remains a critical aspect of long-term care in the myelomeningocele population.
32. **e. Associated with voiding dysfunction, which is usually reversible.** Guillain-Barré syndrome is an acute inflammatory disorder thought to be due to some inflammatory process because most patients do have evidence of acute inflammatory conditions

before development of the syndrome. The syndrome is characterized by a distinct presenting scenario (ascending paralysis). Lower urinary tract dysfunction has been reported in as few as 25% to as many as 80% of cases and is usually reversible with resolution of the disorder.

33. **e. Maximum bladder capacity decreased.** One must remember the potential artifact that significant reflux can introduce into urodynamic studies. Measured bladder capacity may be more, and measured pressures at given inflow volumes may be less, than those after reflux correction. The apparent significance of detrusor overactivity may thus be underestimated.
34. **b. The level of the lesion correlates poorly with urodynamic findings.** Myelomeningocele lesions most commonly occur in the lumbosacral region. The incidence of lower urinary tract dysfunction is estimated to be about 90%, and patients with lower urinary tract dysfunction require life-long follow-up with a urologist.
35. **d. Rash on the palms and soles of feet.** Invasion of the sacral dorsal root ganglia and posterior nerve roots with herpes zoster virus may produce urinary retention, detrusor areflexia, or detrusor overactivity. A rash on the palms and soles is associated with syphilis.

Chapter Review

1. In general, complete spinal cord injury above the sacral spinal cord, but below the area of the sympathetic outflow, results in detrusor overactivity, absent sensation below the level of the lesion, smooth sphincter synergia, and striated sphincter dyssynergia.
2. Lesions at or above the spinal cord level of T7 or T8 (the spinal column level of T6) may result in smooth sphincter dyssynergia as well.
3. Neurologic examination is no substitute for urodynamic evaluation in these patients when one is determining risk factors and treatment.

4. A careful initial evaluation and periodic, routine follow-up evaluation must be performed to identify and correct risk factors and potential complications: high-pressure storage, high detrusor leak point pressure, vesicoureteral reflux (VUR), stone formation (lower and upper tracts), and complicating infection, especially in association with reflux.
5. Avoid use of long-term antibiotics to prevent infection in patient performing clean intermittent catheterization (CIC).
6. Ongoing investigations with ventral nerve surgery and use of stem cells may completely change the paradigm of suprasacral cord injury management in the future.
7. Potential risk factors and complications are those previously described, with particular emphasis on storage pressure, which can result in silent upper tract decompensation and deterioration in the absence of VUR.
8. Treatment is usually directed toward producing or maintaining low-pressure storage while circumventing emptying failure with CIC when possible. Pharmacologic and electrical stimulation may be useful in promoting emptying in certain circumstances.